

MULESHOE 19 S, TX, USA

WMO: 747270

Lat: 33.9557N

Lon: 102.7740W

Elev: 1141

StdP: 88.35

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 03054

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.2	-8.8	-18.2	0.9	0.8	-16.1	1.1	2.9	10.5	13.7	9.6	12.9	1.7	N/A	0.491

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.1	37.2	17.3	35.6	17.8	34.3	18.0	21.4	29.7	20.8	29.2	20.3	28.7	4.5	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.4	16.3	23.4	18.7	15.6	23.1	18.0	14.9	22.6	68.2	29.7	65.7	29.1	63.7	28.2	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.2	7.2	-16.2	39.9	3.0	1.0	-18.4	40.6	-20.1	41.2	-21.8	41.7	-24.0	42.5		
(5)			-16.8	22.3	3.5	0.6	-19.3	22.8	-21.3	23.1	-23.3	23.5	-25.9	23.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.7	3.8	5.9	10.5	14.3	19.0	24.7	25.1	24.6	20.8	14.7	8.5	4.0		
	DBStd	8.89	5.12	5.55	4.96	4.54	4.39	2.66	2.25	2.48	3.09	4.53	4.91	5.13		
	HDD10.0	694	199	133	54	14	2	0	0	0	0	13	84	194		
	HDD18.3	2103	451	348	244	134	46	1	0	0	0	126	296	446		
	CDD10.0	2407	7	18	70	142	280	440	467	452	324	160	38	8		
	CDD18.3	771	0	0	1	11	65	191	209	194	87	13	0	0		
	CDH23.3	10599	2	27	152	443	1179	2583	2564	2219	1046	328	53	2		
	CDH26.7	5246	0	3	31	142	556	1451	1374	1161	432	92	5	0		
(14) Wind	WSAvg	3.1	3.1	3.5	3.6	3.9	3.6	3.4	2.7	2.5	2.5	2.7	2.9	3.1		
(15) Precipitation	PrecAvg	456	16	13	29	25	52	57	59	66	59	38	20	19		
	PrecMax	1028	43	59	100	93	154	159	157	161	261	95	99	83		
	PrecMin	206	0	0	0	0	2	2	0	12	0	0	0	0		
	PrecStd	163	16	15	27	25	44	46	37	38	55	31	22	21		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.2	26.7	29.7	32.4	36.4	39.3	38.5	37.9	34.8	31.5	27.4	23.0		
		MCWB	9.1	10.6	11.7	12.7	14.5	16.1	18.2	18.2	17.9	14.9	11.6	8.9		
	2%	DB	20.5	23.4	27.2	30.2	34.3	37.6	36.3	35.9	33.0	29.3	25.0	20.0		
		MCWB	7.9	9.2	10.9	11.9	13.7	16.5	18.3	18.7	17.4	14.7	10.8	8.6		
	5%	DB	17.7	20.9	24.9	28.2	32.4	35.9	34.7	34.5	31.3	27.2	22.3	17.5		
		MCWB	6.6	8.5	10.4	11.7	13.6	17.1	18.9	18.9	17.4	14.3	9.9	7.4		
	10%	DB	14.8	17.9	22.2	26.0	30.0	34.1	33.3	32.8	29.2	24.8	19.3	14.3		
		MCWB	5.5	7.2	9.3	11.6	13.8	17.8	19.1	19.1	17.6	13.6	8.9	6.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.0	12.0	15.1	16.3	19.9	21.6	22.7	21.9	21.2	20.0	14.1	11.3		
		MCDB	20.4	21.6	22.6	23.4	25.6	31.9	29.0	30.1	28.0	24.5	19.6	16.4		
	2%	WB	8.6	10.4	13.0	15.0	18.5	20.7	21.7	21.3	20.3	18.6	12.7	9.6		
		MCDB	18.9	22.2	20.1	21.2	25.5	29.9	29.4	29.7	27.2	22.3	18.1	16.7		
	5%	WB	7.1	9.0	11.6	13.8	17.5	20.1	20.8	20.7	19.7	17.0	11.2	8.2		
		MCDB	16.8	20.6	21.7	22.4	24.8	29.4	29.2	29.3	26.7	22.3	19.3	16.4		
	10%	WB	5.8	7.6	10.4	12.6	16.4	19.5	20.2	20.1	19.1	15.5	9.9	6.7		
		MCDB	14.2	17.9	21.2	23.1	24.4	28.8	28.8	28.9	25.5	21.4	18.6	14.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.4	17.9	18.2	18.3	17.1	17.0	15.1	14.9	14.8	16.5	17.9	16.4		
		MCDBR	22.2	23.2	22.4	22.4	21.1	20.1	17.5	17.6	18.4	20.3	22.5	21.3		
	5% WB	MCWBR	11.7	11.8	10.3	9.3	8.4	5.9	4.8	5.1	6.3	8.3	10.8	11.6		
		MCWBR	20.2	22.6	20.2	18.5	16.7	15.9	14.4	14.7	13.5	13.9	18.5	18.5		
(40) Clear-Sky Solar Irradiance	taub		0.254	0.263	0.293	0.321	0.333	0.374	0.380	0.368	0.339	0.308	0.284	0.263		
		taud	2.590	2.574	2.491	2.406	2.436	2.367	2.396	2.401	2.455	2.515	2.512	2.551		
	Ebn at Noon		982	1002	987	965	950	907	899	907	923	932	930	946		
		Edh at Noon	76	86	102	117	115	123	119	116	105	91	82	75		
(44) All-Sky Solar Radiation	RadAvg		3.38	4.23	5.48	6.76	7.36	7.73	7.35	6.70	5.64	4.47	3.53	2.93		
	RadStd		0.29	0.34	0.39	0.42	0.49	0.40	0.33	0.30	0.37	0.45	0.33	0.20		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (11 neighbors)	+0.29	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+95	N/A

Nomenclature: See separate page